Md. Minhazul Haq

House 10, Section-11/C, Lane-4, Avenue-5, Dhaka-1216, Bangladesh
mminhaz.haq@gmail.com +8801747202826 GitHub Portfolio
LinkedIn Google Scholar

Research Interests

Machine Learning, Deep Learning, Computer Vision, Autonomous Systems, Intelligent Transportation Systems, Embedded AI, and Agentic AI

Education

Master of Engineering Science (Research) Graduated: July 2026
Department of Electrical Engineering, Universiti Malaya, Malaysia
Field of Research: **Artificial Intelligence**
Course: Research Methodology, 3 credits, Grade A
Thesis: *Optimization of Lightweight YOLO Model for Multiclass Object Detection on Embedded Device*

B.Sc. in Electrical and Electronics Engineering Graduated: Apr. 2022
Jashore University of Science and Technology (JUST), Bangladesh
Final Year **GPA: 3.55/4.00** | **CGPA:** 3.11/4.00 (First Class Honours)
Relevant Coursework: Computer Programming, Digital Signal Processing, Digital Image Processing, Microprocessor and Embedded Systems, Numerical Methods and Statistics in Engineering, Signals Systems.
Thesis: *Design of Solar Tracking System for Optimal Power Generation at Residential Area in Bangladesh*

English Proficiency Test Scores

IELTS: Overall-7.0 (L 7.5, S 7.5, R 7.0, W 6.0) **GRE: 298** (Q 156, V 142, AWA 3.0)

Research and Teaching Experiences

Graduate Research Assistant Nov. 2023 – Feb. 2026
Centre of Intelligent Systems for Emerging Technology (CISET), Universiti Malaya

- **Embedded Object Detection:** Designed and implemented computer vision models for multiclass object detection using natural and UAV image datasets on Jetson Nano.
- **AI for Transportation (Autonomous Perception):** Developed a lightweight vehicle detection and distance estimation system using the Waymo Open Dataset, integrating a modified YOLO26n backbone for vehicle detection with a feed-forward regression network for distance estimation.
- **PPG Stress Monitoring:** Developed a transformer and residual encoder integrated 1D UNet deep-learning model for light stress detection using CASTA and WESAD dataset on Raspberry Pi 3B+ and Arduino Nano BLE platforms.
- **Medical Image Analysis:** Developed attention-enhanced lightweight YOLOv8n models for pediatric wrist fracture and oral cancer detection on resource-constrained devices.
- **AR Application for Engineering Students:** Conducted literature review, questionnaire design, data collection and analysis, and evaluated the Augmented Reality Applications Attitude Scale (ARAAS) to measure attitude toward AR of the electrical engineering students at University Malaya.

Graduate Teaching Assistant / Lab Demonstrator Oct. 2025 – Feb. 2026
LAB 2 (KIE1002), Department of Electrical Engineering, Universiti Malaya

- Students were briefed prior to laboratory sessions, closely monitored experiments, assisted with troubleshooting, and evaluated laboratory performance in **electrical and electronic measurements, circuit analysis, instrumentation, and robotics laboratories**.

Publications

1. **M. M. Haq**, A. S. M. Khairuddin, E. Hanafi, H. M. F. Noman, M. H. Junos, "*HAT-YOLO: Hybrid Attention and Transformer-Integrated Model for UAV Object Detection on Embedded Devices*," *KSII Transactions on Internet and Information Systems*, vol. 20, no. 2, pp. 920-941, 2026. *DOI: 10.3837/tiis.2026.02.014*
2. Yang Z, Khairuddin AS, Huang CJ, Ru WW, Noman HM, Putri TW, Xu X, **Haq MM**, "*A Review of Short-Term, Long-Term, and Memory Consolidation Mechanisms in the Hippocampus and Cerebral Cortex*," *IEEE Access*, 2025. *DOI: 10.1109/ACCESS.2025.3555539*
3. **M. M. Haq**, ASM Khairuddin, and E. Hanafi, "*EdgeLite-YOLOv8n: An Efficient YOLOv8n Model for Real-Time Vehicle Detection in UAV Imagery on Embedded Platforms*," Accepted at the 2nd International Transdisciplinary Research Conference 2026, Universiti Malaya.
4. **M. M. Haq**, ASM Khairuddin, and E. Hanafi, "*An Optimized YOLOv8n Model for Real-Time Small-Object*

Detection on Resource-Constrained Platforms," Review in progress, *Multimedia Tools and Applications*, Submission ID: MTAP-D-26-02236.

5. **M. M. Haq**, ASM Khairuddin, and E. Hanafi, "*Towards Real-Time Wood Defect Detection on Resource-Constrained Devices: An Optimized YOLOv8n Model*," Review in progress, *Forest Science*, Submission ID: FRSC-D-26-00118.
6. Hasan, M. M., K, S., A, M., & **Haq, M. M.**, "*Development of Prototype Dual-Axis Solar Tracking Device for Optimal Power Generation at Residential Area in Bangladesh*," *Journal of Energy and Safety Technology*, 7(1):25–33, 2024.

Conference & Poster Presentation

1. **M. M. Haq**, M. O. Faruk, A. S. B. M. Khairuddin, Effariza Hanafi, K. L. Pua, and W. M. Tilakaratne, "*Real-Time Oral Cancer Detection Using Lightweight YOLOv8n-Based Deep Learning and Computer Vision: A Pilot Study*," UMHEALTH Academic Month & Research Carnival 2025, Universiti Malaya.
2. **M.M. Haq**, "*Lightweight YOLOv8 with Attention-Enhanced Backbone for Real-Time Paediatric Wrist Fracture Detection Using Resource-Constrained Devices*," 4th International Orthopaedic Postgraduate Symposium (IOPS4), Universiti Malaya, 29 Oct. 2025. **Postgraduate Oral Presentation – Winner.**

Projects

Master's Research Projects

- **Lightweight UAV Object Detection and Embedded AI** – Developed lightweight YOLO-based models for real-time object detection in UAV imagery and evaluated their performance on GPU- and resource-constrained embedded platforms.
- **Stress Detection from PPG Signals** – Developed deep-learning models for stress detection from photoplethysmography (PPG) signals using CNN, UNet, residual and transformer encoder based architectures.

Undergraduate Projects

- **Dual-Axis Solar Tracking System** – Designed a dual-axis solar tracking system for optimal power generation in residential applications.
- **Voice-Controlled Humanoid Robot** – Developed a microcontroller-based humanoid robot with voice-controlled movement and interactive functions.
- **Solar Scooter** – Developed a rechargeable electric solar scooter capable of charging through both AC power and solar panel.

Technical Skills

Programming: Python, R, C, C++, MATLAB
AI/ML Packages: Numpy, PyTorch, Pandas, Scikit-learn, Matplotlib, tensorflow-keras, Seaborn, Ultralytics
Simulink Softwares: Simulink, Pspice, CST, Proteus
Hardware: Jetson Nano, Arduino, Raspberry Pi
Documentation Softwares: MS Office, Latex, Adobe Photoshop, Illustrator, Inkscape

Awards & Achievements

- **Best Presenter (Postgraduate Oral Presentation) – Winner**, 4th International Orthopaedic Postgraduate Symposium (IOPS4), Universiti Malaya, 2025.
- **Awarded Top 75 Startup Student to Startup Chapter-2:** Selected among the Top 75 startups in the national competition that organized under the ICT Division, Bangladesh, 2019.

Extra-curricular Activities

- **Journal Reviewer**, *KSII Transactions on Internet and Information Systems*, 2026.
- Organizer, Seminar on **Artificial Intelligence and Machine Learning**, Jashore University of Science and Technology, 2020.
- Organizing Secretary, JUST Robo Society, 2020–2021.

References

Assoc. Prof. Dr. Anis Salwa Binti Mohd Khairuddin
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Assoc. Prof. Dr. Effariza Hanafi
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